Candidate Number ______________ Fraction ______________

The **31st** Republic of China Biology Olympiad National Team Selection Camp **2020**

Experiment **3 :** Botany

Part 1: Investigating the effects of different hormones on Arabica mustard seedlings using histochemical **GUS** staining

Gene expression. Materials and experimental equipment:

A. Experimental plant materials:		quantity
1. Arabian mustard seedling A (transgenic seedling with the promoter of the A gene fused to the GUS reporter gene; PA::GUS) was tested in darkness or under white light (light intensity 80 μmol/m² s) in groups containing different hormones (control group). Grow for 6 days in a medium containing hormone-free (50 μM MeJA or 0.5 μM ACC).		10 each Plant/each people
B. Experimental Equipment:	Quantity C of drug solution:	quantity
1. Dissecting microscope	1 unit of fixative (placed on ice)	5ml
2. Micropipettes P1000	1 vial of 20ml NaPO4 buffer (place on ice)	
3. Micropipette tip	1 box of 3 GUS staining solution (place on ice)	2ml
4. Tweezers (shared with Experiment 2)	2 bottles of distilled water	45ml
5. 1.5 ml microcentrifuge tubes	15 5.50% alcohol	10ml
6. Cover slide/slip (shared with Experiment 2)	7 groups of 6.70% alcohol	15ml
7. Hand towels and toilet paper (shared with Experiment 2) 1 set	7. 95% alcohol	10ml
8. Oil-based ballpoint pen	1 D, shared instrument	
9. Waste liquid cup	1. Plant cultivation box	1 unit
10. Trash can (for throwing away tips, etc.)	1. Vacuum machine	2 units
11. Timer	1. A 3-dimensional photographic dissecting microscope	3 units

Part Two: Seedling Structure, using soybean sprouts as an example. Materials and Experimental Equipment:

1. Soybean sprouts	3 shares
2. Single-sided blade	1 piece
3. Petri dish	1
4. Fine tweezers	2 (shared with Experiment 1)
5. Cover/slide	7 groups (shared with Experiment 1)
6. Dye (0.25% Safranin O in water) 1 small dropper bottle	
7. Hand towels and toilet paper	Group 1 (shared with Experiment 1)
8. A set of duplex microscope + electronic eyepiece + laptop	

Please note:

1. The materials and equipment on the table will not be replenished once they are used up.
2. This exam paper (including cover and question paper) consists of 6 pages and must be returned in its entirety upon submission.
3. The total time allotted for answering questions in this exam is 90 minutes.
4. Please answer the questions on this paper. You may write your answers on the back, but please indicate the question number.

Note: Operation time is limited, please make good use of the time waiting for the experiment to take effect.

Part 1: Investigating the effect of different hormones on gene expression in Arabica seedlings using histochemical **GUS** staining **(50%):** This practical exercise involved treating Arabica seedlings with different hormones (methyljasmonic acid MeJA and ethylene precursor ACC).

Mustard seedlings A were cultured in darkness or white light for 6 days, and then subjected to histochemical GUS staining to obtain...

To understand the effects of these hormones on the expression of the A gene.

II. Experimental Methods, Results, and Discussion:

Experimental Methods:

A. Histochemical **GUS** staining of Arabian mustard seedlings:

1. Six species of Arabica mustard, each containing different hormones and a relative control group, were sampled, with three plants from each species grown in either darkness or light.

   The seedlings were placed into 1.5 ml microcentrifuge tubes and labeled (examination number - white/black night - hormone).

   Record it.

2. Add an appropriate amount of pre-cooled 0.5 ml fixative. After the seedling tissue is completely soaked, open the cap and remove the contents.

   Use a vacuum chamber, then evacuate the air and let it stand at room temperature for 10 minutes.

3. Then rinse the seedlings twice with 0.5 ml of pre-cooled $NaPO_4$ buffer.

4. Add 0.3 ml of pre-cooled GUS staining solution, enough to completely soak the seedlings.

5. To facilitate staining of seedlings, open the caps of the microcentrifuge tubes and place them in a public vacuum chamber for 5 minutes to remove air.

   After 35 minutes, place the lid on the plant culture box in the 37°C common area and gently shake it for 35 minutes.

6. Wash the Arabian mustard seedlings exposed to white light with distilled water to remove the dye, then add 1 ml of 50% [a specific solution] in batches.

   Soak and rinse in 70%, 95%, and 95% alcohol solutions, respectively, for 3, 3, 5, and 5 minutes each.

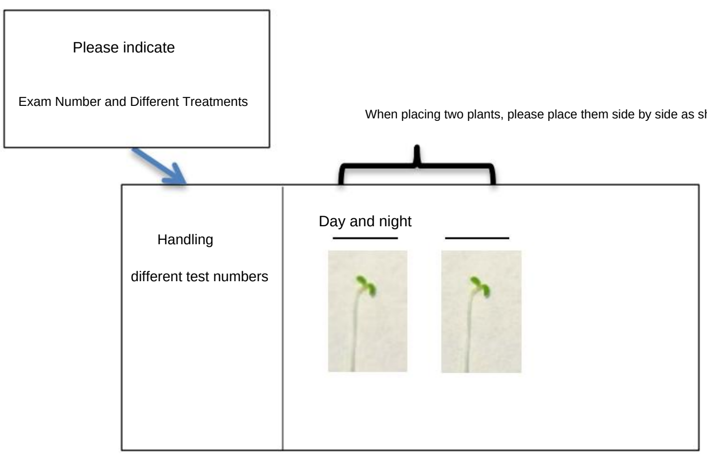
   Gradually replace it with a higher concentration of alcohol to decolorize the chlorophyll in the leaves.

7. Finally, place in a centrifuge tube containing 0.5 ml of 70% alcohol for preservation.

8. Take one seedling from each plant and prepare three water-embedded slides. Examine the staining and gene expression under a microscope.

   The current procedure is to take the slide to the public dissecting microscope photography area and ask the teaching assistant to take a picture, then return to your seat to look at the slide and answer the questions.

When placing two plants, please place them side by side as shown in the picture below to facilitate taking photos.



B. Results and Discussion:

1. Based on the recorded results, describe the effects of different stimuli on Arabica mustard seedlings *(PA::GUS)* in darkness or white light.

The effects of methylated jasmonic acid (MeJA) or ethylene induction are primarily observed in which/which locations of the seedlings?

Does it display color? **(12 points)**

I. In the darkness:

Jasmonic acid: ____

Ethylene: ___________________________

II. In white light:

Jasmonic acid: ____

Ethylene: ___________________________

2. Do these two different hormones share any common location in which the A gene is expressed in darkness and white light?

where? **(6 points)**

I. In the darkness: ______________________________

II. In white light: ______________________________

3. Is the expression of gene A regulated by light? Explain. **(8 points)**

4. Does the expression of the A gene by ethylene differ between light and darkness? Explain. **(8 points)**

5. Infer whether methylated jasmonic acid and auxin differ in their location of induction of A gene expression. Explain.

Its physiological significance. **(8 points)**

Part Two: Seedling Structure, using soybean sprouts as an example **(50%):**

This practical exercise aims to understand the structure of seedlings, especially the distribution pattern of vascular bundles, through freehand sectioning and staining methods.

Mode.

II. Materials and Equipment (per person)

1. Soybean sprouts	3 shares
2. Single-sided blade	1 piece
3. Petri dish	1
4. Fine tweezers	2 (shared with Experiment 1)
5. Cover/slide	7 groups (shared with Experiment 1)
6. Dye (0.25% Safranin O in water) 1 small dropper bottle	
7. Hand towels and toilet paper	Group 1 (shared with Experiment 1)
8. One duplex microscope + one electronic eyepiece + one laptop.	

III. Operating Procedures

1. Slice the different parts of the soybean sprouts by hand (horizontal or vertical slices), and place the slices in a container...

   In a petri dish containing water

2. Select the best sections, stain them, prepare water-embedded slides, observe them under a microscope, and record the results.

   File names for photographs taken with an electronic eyepiece: Exam Number - Section Location. Then, please record the photographs on this sheet of paper.

   The photos taken (please refer to the file name when recording).

IV. Results Record

1. Draw a simplified diagram of a soybean sprout, showing the root, hypocotyl, cotyledons, and young bud of the entire plant.

   Vascular bundle structure and connection patterns, with the structure name and corresponding photograph of its slice labeled in the figure.

   File name.

Scoring method: Simplified diagram *10%* Those slices photographed with clear results and whose structure corresponds correctly to the simplified diagram of bean sprouts; have to *10* point *(10*4* Location, *40%)* Total score *50* point.